: INCH-POUND :

MIL-P-51493(EA)
AMENDMENT 3
6 June 1990
SUPERSEDING
AMENDMENT 2
31 August 1988

MILITARY SPECIFICATION

PAPER, CHEMICAL AGENT DETECTOR: M9

This amendment forms a part of MIL-P-51493(EA), dated 8 April 1983, and is approved for use within Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1.1, under "SPECIFICATIONS": Insert the following:

"COMMERCIAL ITEM DESCRIPTION

A-A-42 - Talcum Powder."

PAGE 4

3.1.1.1(a), line 1, after "hardwood sulfate": Insert ", or a mixture of one-third bleached softwood sulphate, one-third bleached hardwood sulphate, and one-third bleached softwood sulphite".

3.1.1.1(b), line 3, after "colloid": Insert "or dispersion".

3.1.1.1(e), line 2: Delete "or equivalent." and substitute "or "Paramel HE2", or equivalent."

PAGE 5

Table I, under "Minimum", for "Caliper (thickness), millimeters": Delete "0.140" and substitute "0.11".

3.1.2.1(a): Delete in its entirety and substitute the following:

"(a) Mylar. The mylar shall be a matte mylar, DuPont 92EB11; or a semi-matte mylar, DuPont 92S or DuPont 92A; or a smooth mylar, DuPont 92D or equal. Thickness of mylar sheet shall fall in the range 0.8 to 1.1 mils."

AMSC N/A FSC 6665

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3.1.2.1(b), line 2, after "H. B. Fuller SC1204": Insert "or Martac A8416".

3.1.2.1(c): Delete in its entirety and substitute the following:

"(c) Release liner. The release liner shall be two-side silicone coated release liner, Mactac Canada 70 pound (lb) Chocolate Release, GLASSIN Canada 70 lb Chocolate Century or equal. It shall be nominally 70 lb/3000 square feet ream super calendared kraft silicone release coated on both sides. Thickness of the release liner shall fall in the range 3.5 to 4.0 mils. The release liner shall be brown color no lighter than Pantone Matching System (PMS) 464U or 464C or equal. No more than 6 months shall have passed from the date of manufacture of the release liner to the date of manufacture of the backing."

3.1.2.2, lines 4 and 5: Delete last sentence and substitute "For the adhesive B layer, a border greater than or equal to 1/32 inch and less than or equal to 1/4 inch containing no adhesive on either edge is permitted."

3.1.2.3: Delete in its entirety.

PAGE 7

Table II: Delete in its entirety.

3.3, Table III, under "MAXIMUM" column, opposite "Peel strength from steel, OZ per inch": Delete "128" and substitute "160".

PAGE 8

3.7: Delete in its entirety and substitute the following:

"3.7 Workmanship. M9 paper and dispenser box shall meet the following requirements when tested in accordance with 4.4.4.3.1(d).

(a) The roll of detector paper shall be uniform in color when viewed from a distance of 5 feet, with no specks, spots or light or dark patches.

(b) The edge of the roll of detector paper shall be smooth (that is, free of rough or uneven areas).

(c) The roll of detector paper shall be free of grease, oil, dirt and other foreign matter.

(d) All surfaces of the detector paper roll shall be free of thin spots, holes, cuts or tears.

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(e) The B1 paper surface of the detector paper roll shall be smooth and have no more than 5 total surface defects (within the first 25 feet of the detector paper roll) of the following types: wrinkles, tunnels, and creases (for example: 1 wrinkle, 2 tunnels and 2 creases) (see 6.3.1).

(f) All surfaces of the detector paper roll shall not have any surface defects severe enough to cause scuffing or problems in dispensing the laminated paper from the dispenser box.

(g) The dispenser box shall be free from damage and grease, oil, dirt and other foreign matter."

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4.3.2.2(b): Delete in its entirety.

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Table V: Delete in its entirety.

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4.4.3.1.2(a): Add the following major defects and note:

"106 Mylar thickness incorrect* (3.1.2.1).
107 Release liner thickness incorrect (3.1.2.1).
108 M9 roll diameter incorrect (5-77-269).

*May be checked prior to assembly."

PAGE 14

4.4.3.2(b): Delete in its entirety.

4.4.4.1.5(a), line 1: Delete "1.00" and substitute "0.10".

PAGES 15 AND 16

4.4.4.2: Delete in its entirety.

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4.4.4.3.1(d): Add "by visual inspection" after "determine conformance to 3.7".

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4.4.4.3.5: Delete in its entirety and substitute the following:

"4.4.4.3.5 Peel strength, M9 paper. Randomly take eight rolls from the sample and test one specimen from each roll for peel strength. More than two failures shall be cause for rejection. Determine peel strength in accordance with PSTC-1 with the following exceptions:

(a) Mount the M9 detector paper specimens on steel panels and condition specimens for 24 hours at 23° ± 2°C and 50 ± 2 percent relative humidity.

(b) After conditioning, place the mounted paper test strips which are to be tested on a flat horizontal surface.

(c) Sprinkle the paper strips with talcum powder conforming to A-A-42. The entire surface of the paper test strip should be white with talcum powder and no green M9 color showing.

(d) With a fine flexible 1/2-1 inch brush, gently brush the excess talcum powder from the paper test strip. The surface of the test strip should retain a uniform light coating of talcum powder; it should have a "dusty" appearance.

(e) The mounted paper test strip is now ready for the completion of the peel adhesion test in accordance with PSTC-1 except that 5.3 of PSTC-1 shall not apply.

NOTE: If during the execution of the peel adhesion test delamination is still found to occur, discard the sample in question and make sure that subsequent samples have slightly more talcum powder coated on them. This should completely prevent delamination."

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6.3: Add the following new subparagraph in sequence:

"6.3.1 Workmanship definition. The M9 chemical agent detector paper roll may exhibit creases, wrinkles and tunnels. These surface defects are caused by poor workmanship. Surface defects develop due to improper lamination or separation of the paper from the backing.

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(a) A tunnel is any separation or air space that exists between the paper and the backing.

(b) A crease is any overlap, fold or seam on the paper surface.

(c) A wrinkle is any ridge or furrow on the paper surface.

Tunnels, creases and wrinkles may exist singularly or in combination with no specific directional orientation or consistency in length or width."

6.9: Add the following new paragraph in sequence:

"6.9 Manufacturing history. Satisfactory results have been obtained in manufacturing the backing by insuring the backing components conform to the physical characteristics given in table VI.

TABLE VI. Physical characteristics of backing

Characteristic	Requirements	
	Minimum	Maximum
Backing thickness, mils	6.9	8.1
Adhesive thickness, dry, mils:		
A-layer	0.9	1.1
B-layer	1.7	1.9
Mylar thickness, mils	0.8	1.1
Release liner thickness, mils	3.5	4.0

"

Custodian:

Army - EA

Preparing activity:

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